26900159 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (Honours)/ MGU-BBA (Honours)/ MGU-BCA (Honours)

REGULAR EXAMINATION October 2026

Fifth SEMESTER

Discipline Specific Core Course (DSC) - MG5DSCCFS300 - DATABASE MANAGEMENT SYSTEM AND SECURITY

(2024 ADMISSION ONWARDS)

Duration: 1 hour 30 minutes **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Very Short Answer Questions

Answer any **10** questions

Each question carries **1** marks

1. Explain the concept of physical data independence [U] / [CO1]
2. Explain the concept of subclass and super classes in EER model. [U] / [CO1]
3. Differentiate between a primary key and a candidate key. [U] / [CO1]
4. Classify different levels of data abstraction. [U] / [CO1]
5. Examine the various file organization techniques used in DBMS. [An] / [CO2]
6. Outline the purpose of an index in DBMS. [An] / [CO2]
7. Examine the meaning of the symbol | in TRC. [An] / [CO2]
8. Illustrate the meaning of the symbol $\wedge$ in DRC. [An] / [CO2]
9. Illustrate one security measure used to prevent unauthorised database access? [A] / [CO3]
10. Classify the security levels used in a multilevel security system? [A] / [CO3]

11.	Employ a log record to cover an update made by a concurrent transaction?	[A] / [CO3]
12.	Identify the storage types used for maintaining recovery information.	[A] / [CO3]

[1x10 = 10]

Part B

Short Answer Questions

Answer any **4** questions

Each question carries **5** marks

13.	Distinguish between entity integrity constraints and referential integrity constraints.	[U] / [CO1]
14.	Classify the symbols used in ER Model to distinguish between simple, composite, multivalued , derived and key attributes.	[U] / [CO1]
15.	Outline the applications of sequential file organization with suitable examples.	[An] / [CO2]
16.	Examine the different DML statements in SQL and explain their purposes with examples.	[An] / [CO2]
17.	Classify valid and invalid foreign-key values in a student relation and show their effect on referential integrity.	[A] / [CO3]
18.	Use GRANT options to construct a privilege-propagation scenario involving users A1,A2 and A3?	[A] / [CO3]

[5x4 = 20]

Part C

Essay Questions

Answer any **2** questions

Each question carries **10** marks

19.	Identify the major limitations of traditional file processing systems	[U] / [CO1]
20.	Differentiate aggregate functions and sort functions. Illustrate the use of aggregate functions and ORDER BY with suitable SQL statements.	[An] / [CO2]
21.	Implement a checkpoint in a given transaction log and illustrate how it reduces recovery work.	[A] / [CO3]

[10x2 = 20]